

Discrete theory for a mass-conservative adaptive FAS multigrid solver for cell-centered finite differences: solvability, conservation, and second-order convergence for $-\nabla \cdot (D \nabla u) + Cu = f$

EllipticBSAM documentation

following W. Feng, Z. Guo, J. S. Lowengrub, S. M. Wise, *J. Comput. Phys.* 352 (2018) 463–497

Abstract

We collect the discrete theory behind the **EllipticBSAM** solver: the composite block-structured, cell-centered discretization of $-\nabla \cdot (D \nabla u) + Cu = f$ on a rectangle with Dirichlet/Neumann data, closed at coarse-fine interfaces by quadratic (QIF) or linear (LIF) ghost interpolation together with flux-balanced zombie variables. We prove, generalizing the constant-reaction Helmholtz results of Feng–Guo–Lowengrub–Wise to variable $C(x, y) \geq 0$ and to physical Dirichlet boundaries: (i) a kernel characterization and diagonal dominance of the composite matrix; (ii) unique solvability; (iii) exact discrete mass conservation, both of the composite discretization and of the mass-corrected coarse-level force used in the adaptive FAS (AFAS) iteration; (iv) the fixed-point property of the modified-force two-grid algorithm; and (v) global second-order accuracy of the QIF scheme by an M -matrix barrier argument, with the corresponding conservative-flux argument for LIF. We also prove that the boundary-folded red-black Gauss–Seidel smoother used in the implementation performs *exact* Gauss–Seidel updates of the ghost-eliminated equations. Numerical verification tables produced by the accompanying test suite are quoted at the end.

1 The composite discretization

1.1 Setting

Let $\Omega = (a, b) \times (c, d)$ and consider

$$-\nabla \cdot (D(x, y) \nabla u) + C(x, y) u = f(x, y) \quad \text{in } \Omega, \quad (1)$$

with $D \in C^3(\overline{\Omega})$, $D \geq D_{\min} > 0$, $C \in C(\overline{\Omega})$, $C \geq 0$, and on each side of $\partial\Omega$ either Dirichlet data $u = g$ or Neumann data $\partial_n u = g$. (The paper’s Helmholtz problem is $C \equiv \beta > 0$, $D = M$, homogeneous Neumann.)

The root level $\ell = 0$ carries a uniform grid of cell size h_0 ; refinement levels $\ell = 1, 2, \dots$ consist of unions of rectangular patches with $h_\ell = h_0 2^{-\ell}$, aligned so that every fine cell pair exactly tiles a parent cell, *properly nested* (each level- $(\ell + 1)$ cell has its full 8-neighborhood of parent cells covered by level ℓ) and *1-conforming* (at most one hanging node per edge, i.e. level jumps of exactly one across any interface). For the analysis it suffices to consider the two-level composite mesh Ω_γ : coarse cells $\Omega_{0 \setminus 1}$ not covered by the fine patch union Ω_1 , plus the fine cells; the multilevel scheme follows by recursion.

On every level the discretization is the conservative flux form

$$C_{ij}u_{ij} + \frac{F_{i+\frac{1}{2},j}^{ew} - F_{i-\frac{1}{2},j}^{ew}}{h} + \frac{F_{i,j+\frac{1}{2}}^{ns} - F_{i,j-\frac{1}{2}}^{ns}}{h} = f_{ij}, \quad F_{i+\frac{1}{2},j}^{ew} = -D_{i+\frac{1}{2},j} \frac{u_{i+1,j} - u_{ij}}{h}, \quad (2)$$

with D evaluated at face midpoints (after regriding, faces covered by finer cells carry the average of their two fine face values, the compatibility condition (4.15) of the paper).

1.2 Ghost and zombie closures

Fine cells adjacent to the coarse-fine interface Γ_{cf} use *ghost* values u^g located outside Ω_1 :

- **QIF**: the quadratic interpolant along the 45° line through the diagonally adjacent coarse cell center and the two interior fine cells on that line (paper eq. (3.12)/Fig. 4a),

$$u^g = \frac{8}{15} u^c + \frac{2}{3} u_1^f - \frac{1}{5} u_2^f, \quad (3)$$

with the tangentially interpolated coarse value $u^c \mapsto \frac{5}{32} u_+^c + \frac{15}{16} u_0^c - \frac{3}{32} u_-^c$ at the two end cells of each face;

- **LIF**: the linear interpolant $u^g = \frac{2}{3} u^c + \frac{2}{3} u_{\text{same}}^f - \frac{1}{3} u_{\text{partner}}^f$ (paper eqs. (4.13)–(4.14)).

Both reproduce constants: $\frac{8}{15} + \frac{2}{3} - \frac{1}{5} = 1$ and $\frac{2}{3} + \frac{2}{3} - \frac{1}{3} = 1$.

Coarse cells adjacent to Γ_{cf} from outside use *zombie* values at the covered neighbor location, determined by *flux balancing* (paper eq. (4.10)): with u^o the outside coarse value, D_c the coarse interface-face diffusivity and $D_{f,1}, D_{f,2}$ the two fine face diffusivities on that coarse face,

$$D_c (Z - u^o) = D_{f,1} (u_{\text{in}}^{f,1} - u^{g,1}) + D_{f,2} (u_{\text{in}}^{f,2} - u^{g,2}), \quad (4)$$

which states exactly that the coarse face flux equals the sum of the two fine face fluxes, $h_0 F_c = h_1 F_{f,1} + h_1 F_{f,2}$. At salient corners of Ω_1 one coarse location carries two zombie values (one per direction); the closure is unchanged.

Physical boundaries use one layer of boundary ghosts: quadratic elimination $u^g = \frac{8}{3}g - 2u_1 + \frac{1}{3}u_2$ for Dirichlet, $u^g = u_1 + hg$ for Neumann.

Substituting (3)–(4) into (2) yields the composite linear system

$$\mathbf{B} \mathbf{u} = \mathbf{f}, \quad \mathbf{B} = \mathbf{A} + \text{diag}(C), \quad (5)$$

where $\mathbf{A}$ collects the diffusion couplings.

2 Structure of the composite matrix

Lemma 2.1 (row sums, signs, kernel). *Let $\mathbf{A}$ be assembled with QIF or LIF closures and Neumann physical boundaries. Then:*

- $\mathbf{A} \mathbf{1} = \mathbf{0}$, i.e. every row sums to zero.
- All off-diagonal entries of $\mathbf{A}$ are ≤ 0 and the diagonal entries are > 0 , unconditionally for LIF, and for QIF provided the diffusivity resolution condition

$$D_{f,\text{second face}} \geq \frac{1}{5} D_{f,\text{interface face}} \quad (6)$$

holds along each interface normal (paper eq. (3.28); automatic for continuous D on sufficiently fine meshes, trivial for constant D).

(iii) $\ker(\mathbf{A}) = \text{span}\{\mathbf{1}\}$.

If a Dirichlet side is present, the rows of boundary cells on that side acquire the additional diagonal contribution $3D_{1/2}/h^2$ and second-neighbor contribution $-D_{1/2}/(3h^2)$, and their row sums equal $(3 - \frac{1}{3})D_{1/2}/h^2 > 0$; the corresponding rows are strictly diagonally dominant.

Proof. (i) Constants are reproduced by both ghost closures and, by (4) with all u equal, $Z = u^\circ$; every flux in (2) then vanishes.

(ii) Interior rows are the classical 5-point stencil. For a fine cell adjacent to the interface, substituting (3) into its interface-face flux gives (1D notation, paper eq. (3.41))

$$h^2[\mathbf{A}u]_1 = -\frac{8}{15}D_{1/2}u^c + (D_{3/2} + \frac{1}{3}D_{1/2})u_1 - (D_{3/2} - \frac{1}{5}D_{1/2})u_2,$$

whose off-diagonals are nonpositive exactly when (6) holds. For a coarse cell adjacent to the interface, substituting the zombie (4) gives (paper eq. (3.42))

$$h_0^2[\mathbf{A}u]_m = -D_-u_{m-1} + (D_- + \frac{16}{15}D_+)u_m - \frac{2}{3}D_+u_1^f - \frac{2}{5}D_+u_2^f,$$

with unconditionally nonpositive off-diagonals. (In 2D the tangential end-cell weights $\frac{5}{32}, \frac{15}{16}, -\frac{3}{32}$ introduce one more small positive/negative pair; summing the two contributing rows preserves the sign pattern under (6), cf. the paper's Sec. 4.2.) The LIF computation is identical with weights $\frac{2}{3}, \frac{2}{3}, -\frac{1}{3}$ for ghosts and $-\frac{1}{3}, \frac{4}{3}$ for zombies and requires no condition.

(iii) Suppose $\mathbf{A}u = 0$. Multiplying row i by u_i and summing, all flux contributions pair up across faces — across Γ_{cf} because (4) balances them — giving $\sum_{\text{faces}} D_F (\delta_F u)^2 / h_F = 0$ in the composite face inner product (paper eq. (3.54)); hence all face differences vanish. Alternatively, as in the paper's Lemma 3.1, the chain of flux identities $0 = F_{0,1/2} = \dots = F_{1,m+1/2} = \dots = 0$ forces u constant along each grid line, and the ghost/zombie relations propagate the constant across Γ_{cf} : e.g. for QIF, a function constant on each side satisfies $\alpha_0 = -\frac{1}{15}\alpha_0 + \frac{2}{3}\alpha_1 + \frac{2}{5}\alpha_1 \Rightarrow \alpha_0 = \alpha_1$. $\square$

Theorem 2.2 (unique solvability). *Let $\mathbf{B} = \mathbf{A} + \text{diag}(C)$ with $C \geq 0$ and assume (6) for QIF. If $C \not\equiv 0$ or at least one side is Dirichlet, then $\mathbf{B}$ is an irreducibly diagonally dominant Z -matrix with positive diagonal, hence a nonsingular M -matrix: (5) has a unique solution and $\mathbf{B}^{-1} \geq 0$ entrywise. If $C \equiv 0$ and all sides are Neumann, then $\ker(\mathbf{B}) = \text{span}\{\mathbf{1}\}$ and (5) is solvable iff the compatibility condition $(f, \mathbf{1})_\chi + \oint D \partial_n u = 0$ holds discretely; the solution is unique up to constants.*

Proof. By Lemma 2.1, $\mathbf{B}$ has row sums $C_i \geq 0$, nonpositive off-diagonals and positive diagonal. The composite stencil graph is connected (irreducibility), and at least one row is strictly dominant by hypothesis. Irreducible diagonal dominance with at least one strict row implies nonsingularity, and a nonsingular irreducible Z -matrix with nonnegative row sums is an M -matrix with $\mathbf{B}^{-1} \geq 0$ [2, Ch. 6]. The singular statement follows from Lemma 2.1(iii) and the Fredholm alternative for the (after diagonal scaling, cf. paper eq. (3.23)) symmetrizable operator. $\square$

3 Discrete mass conservation

Theorem 3.1 (conservation of the composite scheme). *Let $\mathbf{u}$ solve (5) with QIF or LIF closures. Then, with $(v, w)_\chi = \sum_{\text{visible}} h_i^2 v_i w_i$ the composite inner product (paper eq. (3.45)),*

$$(C\mathbf{u}, \mathbf{1})_\chi - \oint_{\partial\Omega} D \partial_n^h u \, ds = (\mathbf{f}, \mathbf{1})_\chi, \quad (7)$$

where $\partial_n^h u$ is the one-sided ghost-based normal difference quotient and the boundary integral is the sum of face values weighted by face length. In particular, for $C \equiv \beta$ and homogeneous Neumann data, $\beta(\mathbf{u}, \mathbf{1})_\chi = (\mathbf{f}, \mathbf{1})_\chi$: the paper’s mass identity.

Proof. Multiply each cell equation (2) by the cell area and sum over all visible cells. Each interior face appears in exactly two cell equations with opposite signs and equal magnitude $h_F F_F$, so all interior coarse-coarse and fine-fine contributions cancel. A coarse-fine interface face appears once in the outside coarse cell equation, with magnitude $h_0 F_c$, and once in each of the two adjacent fine cell equations, with magnitudes $h_1 F_{f,1}, h_1 F_{f,2}$; by the flux balance (4) these cancel as well. What remains is exactly (7). $\square$

Remark 3.2. Equation (7) is verified to relative size $\sim 10^{-16}$ by `eb_mass_check` at the solved state on all multilevel test hierarchies (Sec. 7); replacing the zombie by the plain restriction (the non-conservative “QI” variant) breaks the cancellation and produces defects of size $O(h^2)$ per interface face, observed at $\sim 10^{-5}$ – 10^{-4} .

4 The mass-corrected AFAS iteration

The solver does not assemble $\mathbf{B}$. It iterates the FAS two-grid algorithm (paper alg. **a0**–**a9**, recursively in the multilevel case) in which smoothing, restriction R (4-point average) and prolongation P (piecewise-constant in the analysis, bilinear in the implementation) are entirely standard, and the only nonstandard ingredient is the *coarse force*: on the covered region $f_0|_{0\neq 1} = R(\mathbf{f}_1 - \mathbf{B}_1 \mathbf{u}_1) + \mathbf{B}_0(R\mathbf{u}_1)$ (standard FAS), and at every uncovered coarse cell adjacent to Γ_{cf} the *flux-balance mass correction* (paper eq. (4.20))

$$C^{(\cdot)}(\mathbf{u}) = \frac{D_c(Z(\mathbf{u}) - [R\mathbf{u}_1])}{h_0^2} = \frac{D_{f,1}(u_{\text{in}}^{f,1} - u^{g,1}) + D_{f,2}(u_{\text{in}}^{f,2} - u^{g,2}) + D_c(u^o - [R\mathbf{u}_1])}{h_0^2} \quad (8)$$

is added to f_0 . Adding (8) is algebraically identical to replacing the covered-neighbor value $R\mathbf{u}_1$ by the zombie $Z(\mathbf{u})$ in that cell’s coarse equation; the smoother never needs to know.

Theorem 4.1 (mass of the corrected coarse force). *Assume the physical boundary contributions to the coarse and fine levels agree (e.g. Neumann data, or any data with the ghost conventions of Sec. 1). Then the coarse force built in step **a4** satisfies*

$$h_0^2 \sum_{(i,j) \in \mathcal{I}_0} f_{0,ij}^{(1)} = h_0^2 \sum_{\mathcal{I}_0 \setminus 1} f_{0,ij} + h_1^2 \sum_{\mathcal{I}_1} f_{1,ij} + (\text{boundary terms}) = (\mathbf{f}, \mathbf{1})_\chi + (\text{boundary terms}).$$

Proof. Identical to the paper’s Theorem 4.2 with βu replaced by Cu : summing $R(\mathbf{f}_1 - \mathbf{B}_1 \mathbf{u}_1) + \mathbf{B}_0(R\mathbf{u}_1)$ over the covered region, the $\mathbf{B}$ -sums telescope to interface fluxes (the variable- C diagonal terms cancel between $R(\mathbf{B}_1 \mathbf{u}_1)$ and $\mathbf{B}_0(R\mathbf{u}_1)$ because R is the conservative average and the covered coarse C equals the average of its fine values by the coefficient restriction of Sec. 1); the four flux-balance identities encoded in (8) (paper p. 483) convert the fine interface fluxes into coarse interface fluxes minus the corrections, which cancel the corrections added at the adjacent outside cells. $\square$

Theorem 4.2 (fixed point). *Let $\mathbf{u}$ solve $\mathbf{B}\mathbf{u} = \mathbf{f}$ (in the singular case, the representative with the prescribed mass). Then $\mathbf{u}$ is a fixed point of the modified-force two-grid map, $\mathbf{u} = \text{TG}(\mathbf{u})$, and conversely every fixed point solves the composite system.*

Proof. As in the paper's Theorems 3.11/4.3. If $\mathbf{B}\mathbf{u} = \mathbf{f}$, the pre-smoothing leaves the (already exact) fine equations invariant in residual form $\mathbf{r}_1 = \mathbf{0}$; the coarse force on the covered region becomes $\mathbf{B}_0(R\mathbf{u}_1)$, and at corrected cells $f_0 + C(\mathbf{u})$ equals the coarse equation evaluated with the zombie neighbor. One verifies (paper eq. (3.84)–(3.86), with D variable) that $\mathbf{u}_0^* = R\mathbf{u}_1$ on the covered region and $\mathbf{u}_0^* = \mathbf{u}_0$ outside solves the coarse system exactly, so the correction step adds zero and post-smoothing changes nothing. The converse direction follows by reading the same identities backwards at a stationary iterate. $\square$

5 Second-order convergence

Theorem 5.1 (global $O(h^2)$ accuracy, QIF). *Let $u \in C^4(\bar{\Omega})$ solve (1), let (6) hold, and let $C \neq 0$ or a Dirichlet side be present. Let $\mathbf{u}_h$ solve the composite system (5) on a 1-conforming, properly nested hierarchy with finest spacing h and a uniformly bounded number of levels. Then*

$$\|\mathbf{u}_h - u\|_\infty \leq C^* h^2,$$

with C^* depending on $\|u\|_{C^4}$, $\|D\|_{C^3}$, $D_{\min}$ and the domain, but not on the interface geometry.

Proof. Write the truncation error $\tau := \mathbf{B}u - \mathbf{f}$ evaluated at the exact solution. Away from Γ_{cf} and $\partial\Omega$, Taylor expansion of the flux form gives the classical $|\tau_{ij}| \leq c_1 h^2$. At a fine cell adjacent to Γ_{cf} , the QIF ghost is a quadratic interpolant along a line with nodes at distances $O(h)$, so $|u^g - u(x^g)| \leq c_2 h^3$; the resulting face-flux error is $O(h^2)$ and, after dividing by h , $|\tau| \leq c_3 h$ at those cells. At an uncovered coarse cell adjacent to Γ_{cf} , the zombie closure (4) replaces the coarse flux by the sum of fine fluxes; each fine flux carries an $O(h^2)$ error (difference quotient $O(h^2)$ plus ghost-induced $O(h^2)$), hence again $|\tau| \leq c_4 h$. At Dirichlet cells the quadratic boundary ghost gives $|\tau| \leq c_5 h$; Neumann faces give $|\tau| \leq c_6 h$ likewise. Thus

$$|\tau| \leq ch^2 \quad \text{in the bulk,} \quad |\tau| \leq ch \quad \text{on the strips } \Sigma := \{\text{cells touching } \Gamma_{cf} \cup \partial\Omega\}.$$

By Theorem 2.2, $\mathbf{B}$ is an M -matrix; we conclude with a comparison (barrier) function. Let $d(x)$ denote the signed distance to Γ_{cf} (positive on the coarse side) and set, for the single-interface case,

$$w(x) = \alpha h^2 (K - |d(x)|)_+ + \gamma h^2 \Phi(x),$$

where $\Phi > 0$ solves $-\nabla \cdot (D\nabla\Phi) + C\Phi \geq 1$ classically (e.g. Φ a suitable quadratic) and $K \geq \text{diam}(\Omega)$. The kinked first term is piecewise linear in d with a slope jump $2\alpha h^2$ across Γ_{cf} ; applying the composite operator, the ghost/zombie closures reproduce linear functions up to $O(h)$ errors with the crucial property that the *flux jump* of w contributes $\geq c D_{\min} \alpha h^2 / h = c D_{\min} \alpha h$ to $[\mathbf{B}w]$ at every cell of Σ , while $[\mathbf{B}w] \geq \gamma h^2$ in the bulk and w 's boundary rows are handled by the strict dominance of Lemma 2.1. Choosing α, γ proportional to the constants c_3, c_4, c_1 above yields $\mathbf{B}w \geq |\tau|$ componentwise, and $w \leq (\alpha K + \gamma \|\Phi\|_\infty) h^2$. By the discrete comparison principle ($\mathbf{B}^{-1} \geq 0$), $\|\mathbf{u}_h - u\|_\infty = \|\mathbf{B}^{-1}\tau\|_\infty \leq \|w\|_\infty \leq C^* h^2$. For finitely many interfaces (bounded number of levels) one sums one kinked barrier per interface. $\square$

Proposition 5.2 (LIF). *With LIF ghosts the interface truncation degrades to $|\tau| = O(1)$ on Σ , but it has the conservative structure $\tau|_\Sigma = \text{div}_h(\delta F)$ with face perturbations $|\delta F| = O(h^2)$ supported on interface faces (this is precisely what flux balancing enforces). Testing the error equation with the discrete energy inner product of the paper's Theorems 3.4–3.6 and using summation by parts on the composite mesh, $\|\mathbf{u}_h - u\|_{2,\chi} \leq C(\|\delta F\| + \|\tau_{\text{bulk}}\|) = O(h^2)$. Second-order*

L^∞ convergence is observed numerically (Sec. 7), matching the paper’s Tables 1–4; without flux balancing (interpolated zombies) the scheme loses both the conservation identity (7) and, on multilevel meshes, visible accuracy — as the paper demonstrates in its Table 6 and our ablation reproduces.

6 Convergence of the iteration

Proposition 6.1 (two-grid contraction). *Write the error propagation of the modified-force two-grid map for the linear problem (5) as $\mathbf{e}^{k+1} = \mathbf{S}^{\nu_2} \mathbf{K} \mathbf{S}^{\nu_1} \mathbf{e}^k$, where $\mathbf{S}$ is the red-black Gauss–Seidel error operator and $\mathbf{K} = \mathbf{I} - \mathbf{P} \mathbf{B}_0^{-1} \hat{\mathbf{R}} \mathbf{B}$ is the coarse-grid correction with $\hat{\mathbf{R}}$ the corrected restriction induced by steps **a3–a4** (the corrections (8) make $\hat{\mathbf{R}} \mathbf{B}$ the zombie- closed coarse operator applied to restricted quantities). $\mathbf{S}$ satisfies the smoothing property on the M -matrix $\mathbf{B}$: $\|\mathbf{B} \mathbf{S}^\nu\|_\infty \leq \eta(\nu) h^{-2}$ with $\eta(\nu) \rightarrow 0$, by the standard regular-splitting argument for Gauss–Seidel on irreducibly dominant Z -matrices. If the approximation property $\|\mathbf{B}^{-1} - \mathbf{P} \mathbf{B}_0^{-1} \hat{\mathbf{R}}\|_\infty \leq C_A h^2$ holds, then $\|\mathbf{S}^{\nu_2} \mathbf{K} \mathbf{S}^{\nu_1}\| \leq C_A \eta(\nu) < 1$ for ν large enough, uniformly in h .*

The approximation property on composite meshes is, as in the literature on FAC/MLAT methods, established case by case; we do not prove it here. Instead the implementation *measures* the contraction: the V(3,3) cycle reduces the composite residual by factors 0.01–0.05 per cycle, independent of h and of the number of levels (Sec. 7), consistent with the paper’s reported geometric, h -independent convergence (their Fig. 8, factor $\approx 1/20$).

Lemma 6.2 (exact boundary-folded smoothing). *For a Dirichlet boundary cell with ghost $u^g = \frac{8}{3}g - 2u_1 + \frac{1}{3}u_2$, define the partial ghost $\tilde{u}^g = \frac{8}{3}g + \frac{1}{3}u_2$ and add $2D_{1/2}/h^2$ to the relaxation diagonal. Then the point update*

$$u_1 \leftarrow \frac{F + D_{1/2} \tilde{u}^g / h^2 + (\text{other neighbor terms})}{C_1 + (3D_{1/2} + D_{3/2}) / h^2 + (\text{transverse faces})}$$

solves the ghost-eliminated equation of u_1 exactly, with all neighbors frozen; i.e. the sweep is exact Gauss–Seidel for the eliminated system. The analogous statement holds for Neumann faces with $\tilde{u}^g = hg$ and diagonal increment $-D_{1/2}/h^2$.

Proof. Substitute u^g into the cell equation and collect the u_1 terms: the west-face flux contributes $D_{1/2}(u_1 - u^g)/h^2 = D_{1/2}(3u_1 - \frac{1}{3}u_2 - \frac{8}{3}g)/h^2$, i.e. $+3D_{1/2}/h^2$ to the diagonal and $-(D_{1/2}/3)/h^2$ to the u_2 coupling; the displayed update implements precisely this row. (Without the fold, the $-2u_1$ hidden in a frozen ghost makes the boundary update a damped Jacobi step with factor $\approx \frac{1}{2}$, which measurably degrades the V-cycle: in our tests the cycle count for 10^{-12} dropped from 22 to 10 when the fold was enabled.) $\square$

7 Numerical verification

All numbers below are produced by `tests/run_all_tests.m` (MATLAB R2025b, double precision); see `TEST_REPORT.md` for the complete logs.

Uniform grids. Variable $D = 1 + \frac{1}{2} \sin 2\pi x \cos \pi y$, $C = 1 + xy$, Dirichlet data: L^∞ errors $3.29 \cdot 10^{-4}$, $8.31 \cdot 10^{-5}$, $2.09 \cdot 10^{-5}$, $5.23 \cdot 10^{-6}$ for $m = 32, \dots, 256$; rates 1.98, 1.99, 2.00. Pure-Neumann singular Poisson: rates 1.99, 2.00, 2.00.

Fixed multilevel meshes (paper Tables 2/3 setup). Periodic Gaussian (paper eq. (5.1)) with $C = 1$, exact Neumann data, box-in-box 2/3/4-level hierarchies, $m_0 = 32 \rightarrow 256$:

	$m_0=32$	64	128	256
2-level L^∞	$5.95 \cdot 10^{-3}$	$1.51 \cdot 10^{-3}$	$3.78 \cdot 10^{-4}$	$9.45 \cdot 10^{-5}$
3-level L^∞	$9.10 \cdot 10^{-3}$	$2.29 \cdot 10^{-3}$	$5.72 \cdot 10^{-4}$	$1.43 \cdot 10^{-4}$
4-level L^∞	$4.45 \cdot 10^{-3}$	$1.12 \cdot 10^{-3}$	$2.81 \cdot 10^{-4}$	$7.02 \cdot 10^{-5}$

with all successive rates 1.98–2.00 (QIF; LIF identical to three digits). L-shaped two-patch refinement (paper Table 5 setup, reentrant corner, double zombies): rates 1.79, 2.04, 2.02, 2.01 for $m_0 = 16 \rightarrow 256$.

Conservation. At the solved state, the identity (7) holds to relative size 10^{-16} – 10^{-15} on all hierarchies. Disabling the corrections (8) yields defects $2.3 \cdot 10^{-5}$ (2-level) and $2.9 \cdot 10^{-4}$ (3-level) and degrades the 3-level error from $2.29 \cdot 10^{-3}$ to $1.29 \cdot 10^{-2}$ — the discrete analogue of the paper’s Table 6 comparison.

h -independent geometric solver convergence. Average per-cycle residual reduction factors of the V(3,3) cycle (composite solve from a zero start) on the 2-level mesh, for $m_0 = 32, 64, 128, 256$, are respectively

$$0.045, \quad 0.050, \quad 0.052, \quad 0.053.$$

On 3- and 4-level meshes the factors are 0.05 ± 0.005 , independent of depth. The iteration is geometric and mesh-independent, with factor $\approx 1/20$ exactly as in the paper’s Fig. 8.

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